Unit 14, Lesson 10: Solving One-Step Inequalities Using Multiplication and Division

Benchmark

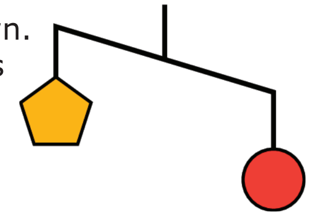
MA.7.AR.2.1 Write and solve one-step inequalities in one variable within a mathematical context and represent solutions algebraically or graphically.

Learning Targets

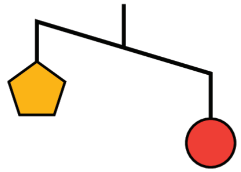
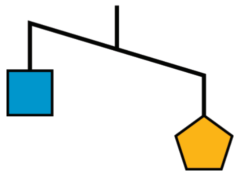
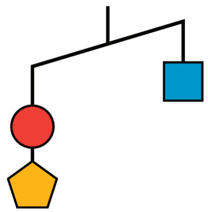
- ☐ I can determine the minimum or maximum value that makes an inequality true.
- ☐ I can explore the relationship between the coefficients of the terms in an inequality and solution set.
- ☐ I can solve one-step inequalities in one variable using multiplication or division.
- ☐ I can relate properties of inequality to solving one-step inequalities algebraically.

14.10.1 Warm-Up: Visual Models

An unbalanced hanger diagram is shown. It shows that the weight of one circle is greater than the weight of one pentagon.



1. Complete the table.

Hanger Diagram	Description	Inequality
	The weight of one circle, c, is greater than the weight of one pentagon, p.	
		$s < p$
	The weight of one square, s, is ___________ than the weight of a circle and a pentagon.	$c + p >$ ___
		$s + c > 2t$

14.10.2 Exploration: Determining the Minimum or Maximum Value that Makes an Inequality True

Work with your partner to complete the following.

1. Bak Middle School of the Arts in West Palm Beach, Florida is hosting a rehearsal for its upcoming musical. The stage manager has $\$83$ from a recent fundraiser that can be used to order lunch for the cast and crew. The school's cafeteria can make boxed lunches for a cost of $\$6$ each. The inequality $6n \leq 83$ represents the situation, where n represents the number of boxed lunches that can be ordered from the cafeteria.

   a. Select the values that represent a valid number of boxed lunches the stage manager can order from the cafeteria.

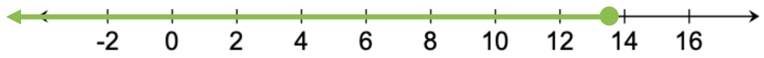
   - ☐ 14
   - ☐ $13\frac{5}{6}$
   - ☐ 12
   - ☐ 9.5
   - ☐ 2
   - ☐ $^{-}4$

   b. Explain why some values in the previous question are not valid in this context.

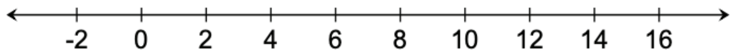
   c. The mathematical solution to the inequality $6n \leq 83$ is $n \leq 13\frac{5}{6}$. What is the maximum number of boxed lunches the stage manager can order from the cafeteria?

   d. What is the minimum number of boxed lunches the stage manager can order from the cafeteria?

e. The graph of the solution set for $6n \leq 83$ is shown.



Discuss with your partner how the graph that mathematically represents the solution set is not the best representation of the real-world context. Then, represent the solutions in the real-world context on the number line shown.



2. Hiromi walks around his neighborhood each afternoon, tracking his pace each day. The fastest rate Hiromi can walk for an extended period of time is 3.7 miles per hour.

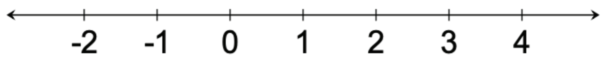
   a. What is the maximum distance Hiromi can walk in 0.75 hours?

   b. The inequality $\frac{d}{3.7} \leq 0.75$ can be used to represent the distance Hiromi walks, d, in 0.75 hours given his fastest walking rate of 3.7 miles per hour.

   Use your answer to Part A to determine the solution set for $\frac{d}{3.7} \leq 0.75$.

   c. The graph of the solution set for $\frac{d}{3.7} \leq 0.75$ is shown.

   Discuss with your partner how the graph that mathematically represents the solution set is not the best representation of the real-world context. Then, represent the solutions in the real-world context on the number line.



14.10.3 Bring It Together: Solutions and Mathematical Models in Context

Each of the inequalities in the Exploration has a mathematical solution set that can be represented on a number line. These graphs are examples of mathematical models.

Mathematical models are helpful tools used to understand a situation and make predictions. However, sometimes the mathematical model can include parts that do not make sense in a real-world context. When interpreting models to solve problems, it is important to analyze whether aspects of the model make sense in a given context.

14.10.4 Collaboration: The Effect of a Negative Coefficient

1. Two simple inequalities are shown.

$$1 > 0 \qquad \text{and} \qquad {}^{-}1 > 0$$

   a. Circle the inequality that is true and ~~cross out~~ the inequality that is false.

   b. Complete the statement.

   If 1 is greater than 0, then the opposite of 1 is

   _______________ than 0.

   c. Write the correct inequalities by filling in a comparison symbol on each line.

$$0 \_\_\_\_\_ 1 \qquad \text{and} \qquad 0 \_\_\_\_ {}^{-}1$$

   d. Verify your answers with your partner on parts A-C and, if necessary, resolve any differences.

2. Work with your partner to complete the following.

   a. Complete the inequalities in the table based on your understanding of opposites.

Given Inequality	Equivalent Inequality
$0 < -q$	0 ___ q
$0 > -m$	0 ___ m
$-w \leq 0$	w ___ 0

In each equivalent inequality, the sign of the variable changed to its opposite and the value 0 remained the same.

   b. Describe which operation(s) could be applied to both sides of the given inequalities above to result in the signs of the variables changing and the value 0 remaining the same.

   c. Describe what happens to the comparison symbol in the inequality when either of these operations are applied.

14.10.5 Bring It Together: The Effect of a Negative Coefficient

1. Complete the statements using words from the bank shown. Some words may not be used.

adding	subtracting	multiplying
dividing	opposite	

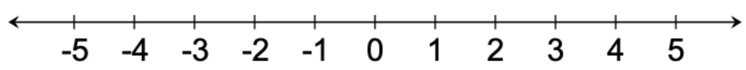
When the variable term in an inequality is negative, the inequality can be rewritten by __________________ or __________________ the inequality by -1. In the resulting inequality, each term and the comparison symbol will change to its __________________.

2. Use the inequality $-n \geq 2$ to complete the following.

   a. Complete the inequality to create an equivalent inequality.

   $n \ \square$ _____

   b. Graph the solution set for $-n \geq 2$.



14.10.6 Guided Instruction: Solving One-Step Inequalities with Multiplication and Division

As with solving one-step inequalities with addition and subtraction, properties of inequality can be used to solve inequalities with multiplication and division. As discovered in the Collaboration, though, when the value that is being multiplied or divided is negative, this changes both the values in the inequality *and* the comparison symbol to their opposites.

Multiplying both sides of an inequality by the same value can be used to solve an inequality by the ***multiplication property of inequality***.

Multiplication Property of Inequality

If $a > b$ and $c > 0$, then $a \times c$ ____ $b \times c$.

If $a > b$ and $c < 0$, then $a \times c$ ____ $b \times c$.

1. Find the solution to both inequalities using the multiplication property of inequality.

$\frac{y}{4} \leq 2.1$	$\frac{y}{-4} \leq 2.1$

Dividing both sides of an inequality by the same value can be used to solve an inequality by the ***division property of inequality***.

Division Property of Inequality

If $a > b$ and $c > 0$, then $a \div c$ ____ $b \div c$.

If $a > b$ and $c < 0$, then $a \div c$ ____ $b \div c$.

2. Find the solution to both inequalities using the division property of inequality.

$-25 > 5c$	$-25 > -5c$

14.10.7 Wrap-Up: Cool Down

Two inequalities are shown.

$$2.5x \geq -10 \qquad\qquad -2.5x \geq -10$$

1. Complete the table to describe how the process of solving each inequality is similar and different.

How is the process of solving each inequality similar?	How is the process of solving each inequality different?

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